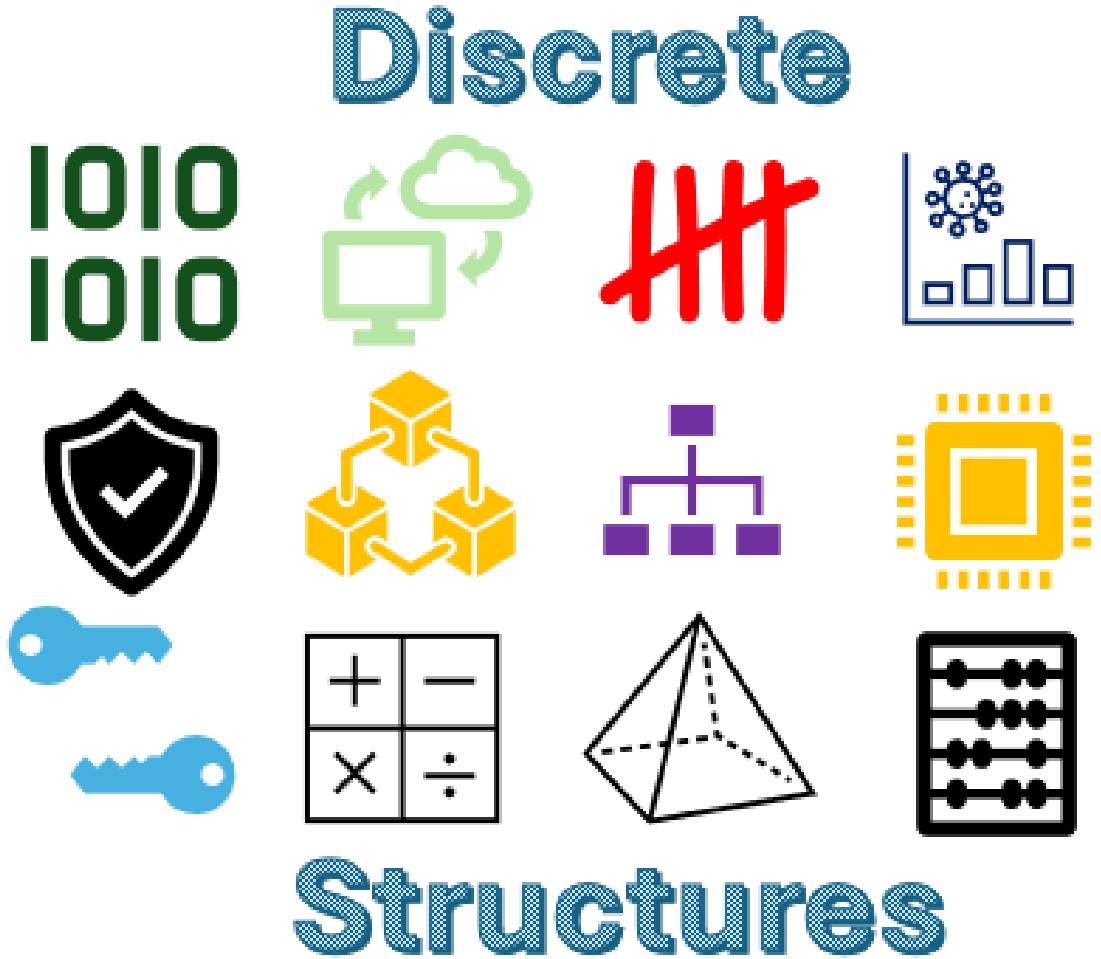
COMP 2310: Discrete Structures
Fall 2026

Lecture 1

i. Course introduction, syllabus discussion, policy, and guidelines

ii. Chapter 1: Fundamentals



About the Instructor

- Suraiya Tairin

- Assistant Professor, Department of Computer Science (starting from Fall 2026)
- PhD in Computer Engineering, University of Virginia

- Office:
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- Email:
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Today’s agenda

- Course syllabus discussion, policy, and guidelines discussion
- Course introduction
- Introduction to general concepts
- Chapter 1: Fundamentals

Course syllabus, policy, and guidelines discussion

About the course

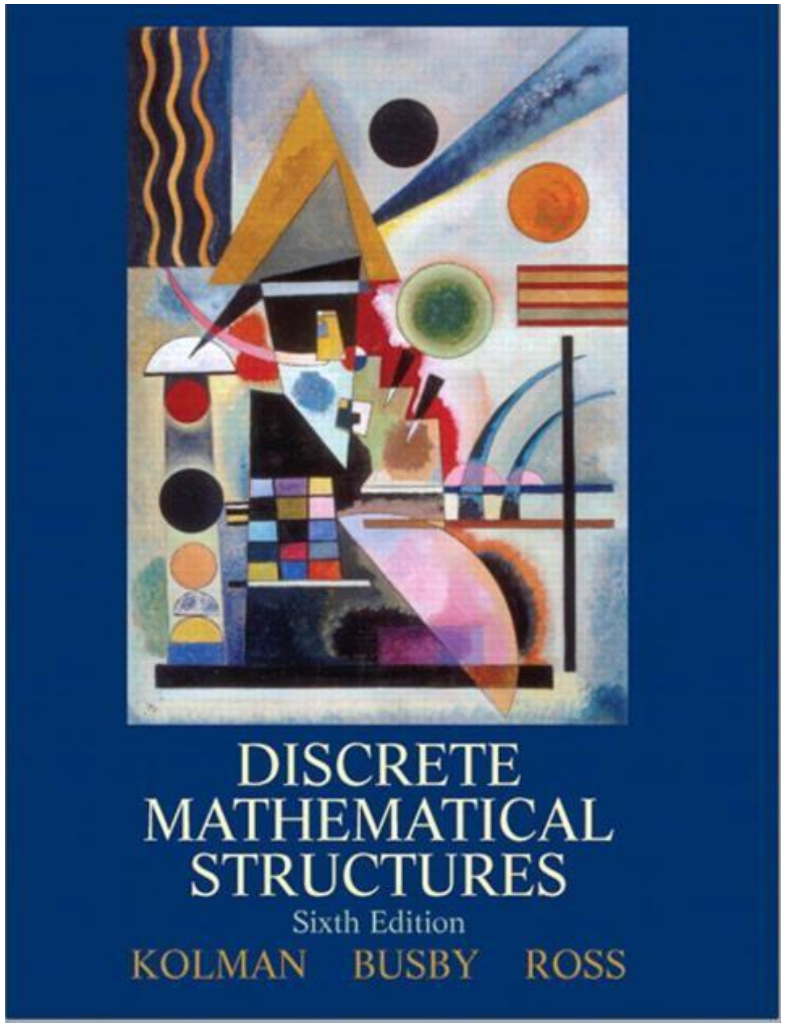
- Course title: Discrete Structures
- Course CRN: 11178
- Credit hours: 3
- Course website (eCourses): https://ecourses.pvamu.edu/courses/41898
- Course meeting: **TR 9.30 am – 10:50 am** at **S.R. Collins Room #226**
- Office hours: TR: 11:30 am - 2 pm. Also available by appointment
- Instructor's office: **S.R. Collins Room #318**

Classroom etiquette

- Please refrain from using any electronic devices unless specifically required for course-related activities (textbook, eCourses etc)
- Please step outside if you need to have a phone conversation
- You are expected to remain in class until the end, as attendance will be recorded toward the end of the session.
- Eating is not allowed in the classroom
- Please try to come to the classroom on time
- Be interactive and engage in discussion as much as possible

Textbook

- Textbook is required: physical copy or digital copy



Discrete Mathematical Structures, 6th edition, ISBN 978-0132297516. Author: Bernard Kolman, Robert C. Busby, Sharon Cutler Ross. Publisher: Pearson Prentice Hall.

Course contents

- **We will cover the following contents from the required textbook**
- **https://catalog.pvamu.edu/universitycourses/comp/**

A bridge course between data structures/discrete mathematics and analysis of algorithms, to include reviews of functions and relations, basic combinatorics (set operations, counting, combinations, and permutations) and introductions to prepositional and predicate logic, discrete probability theory, recursive definitions, computational complexity, and proof techniques including mathematical induction. The concepts are illustrated by applications involving graphs, trees, networks and related algorithms.

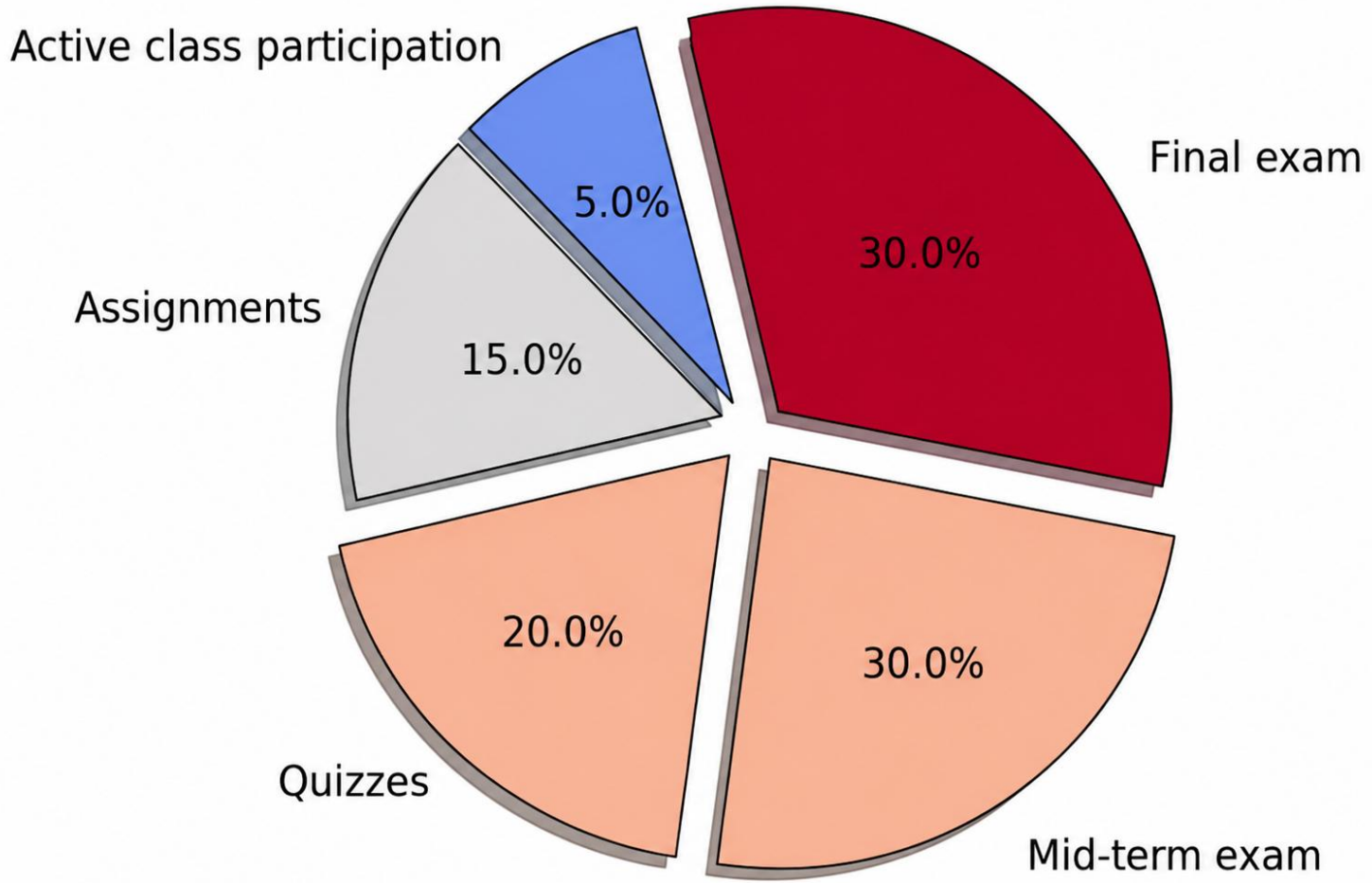
Course outcomes

- Learning outcomes:
  - Demonstrate knowledge and application of discrete mathematics
  - ABET outcome 1
    - Empirical and quantitative skills
  - Demonstrate knowledge and application of computing
  - ABET outcome 1
    - Empirical and quantitative skills

What is ABET

- **ABET: Accreditation Board for Engineering and Technology.**
- Assure the **quality of educational programs** in fields related to engineering, technology, and applied sciences.
- Accreditation signifies that a program meets **established standards** and prepares students for professional practice.
- Graduates from an ABET-accredited program have a **solid educational foundation** and are capable of leading the way in innovation, emerging technologies, and in anticipating the welfare and safety needs of the public.

Mark distribution



Assignments (15%)

- Approximately 5 written assignments
- One week to complete
- Assignments are due at 11:59 pm on the specified date
- May include programming assignment
- Late submissions are accepted up to 5 days after the deadline and penalized by 10% for each day
- No make-up is allowed unless provided University approved excuses
- Will be checked automatically for plagiarism using, please DO NOT copy from any sources.

Quizzes (20%)

- You will be notified at least 24 hours before any quiz
- Based on the materials covered in the previous class
- In-class quiz
- No make-up is allowed unless provided University approved excuses
- 3-5 multiple choice questions
- Approximately 10 quizzes
- No resources including online and offline including book, notes are allowed

Mid-term exam (30%)

- Based on the materials covered from week 1 – 6
- In-person
- No make-up is allowed unless provided University approved excuses
- 30 multiple choice questions
- **Only one page (8.5 x 11 inches, both sides) of handwritten note is allowed.** No other online or offline resources are allowed.

Final exam (30%)

- Based on the materials covered throughout the course
- In-person
- No make-up is allowed unless provided University approved excuses
- 30 multiple choice questions
- **Only one page (8.5 x 11 inches, both sides) of handwritten note is allowed.** No other online or offline resources are allowed.

Attendance

- Attendance is required
- Please do not be late more than 15 minutes unless there is an emergency
- You are expected to stay in the class throughout the entire period
- Your absence may only be excused in the case of unavoidable circumstances

Active class participation (5%)

You are expected to actively participate in activities including but not limited to:

i. analytical discussion
ii. contributing to critical thinking
iii. presenting your ideas and thoughts

Each active participation with **reasonable** discussions/thoughts/ideas is worth 1 point. Therefore, to receive full marks, you must engage at least 5 times throughout the semester.

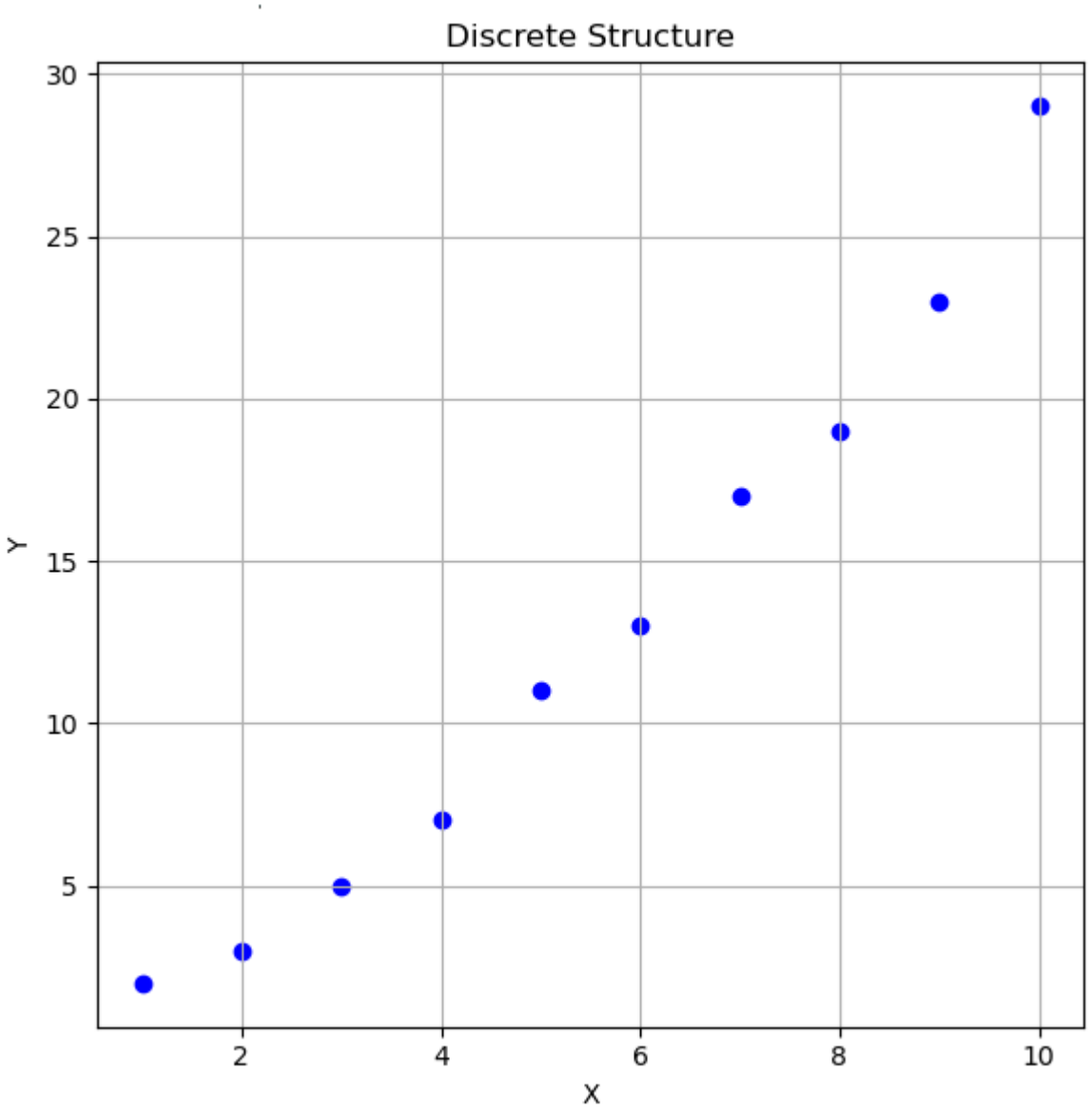
Example: Consider a relation on a set where two elements are related based on a specific condition—what characteristics does this relation have?

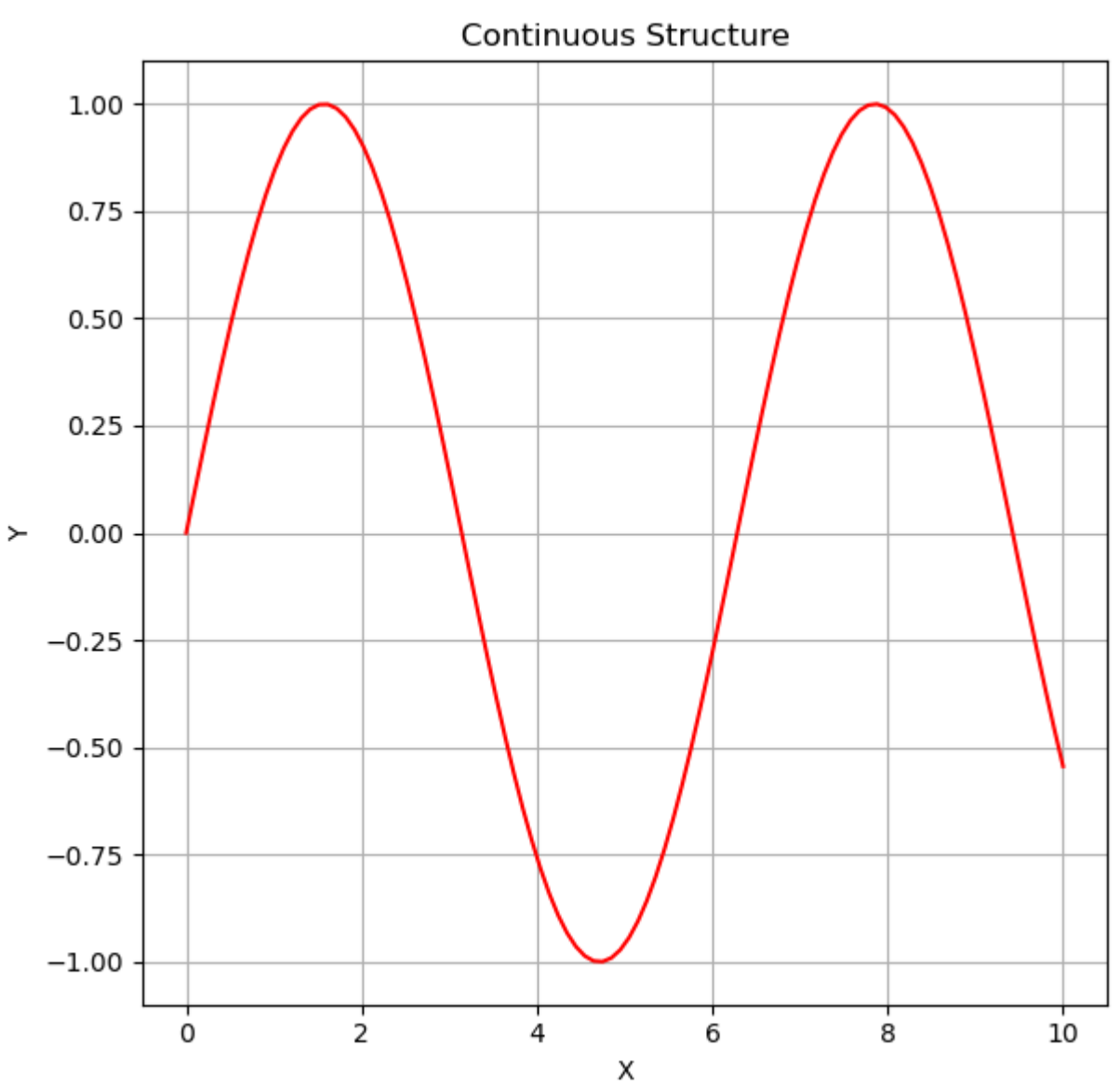
Introduction to the course

What is discrete structures

- **Discrete**
  - Individual, distinct as opposite to continuous (uninterrupted, smooth)
  - **Discrete:** counting objects (1, 2, 3, 4, 5,…) → Each number is different distinct
  - **Continuous:** height (can be of any number within a range → 5 ft – 7 ft)
  - Any other example?
  - discrete data storage vs signal processing
- **Discrete structures:** countable, distinct, separate, and finite structure/elements.
  - Graphs, sets, relations, trees, function, matrix

What is discrete structures (contd.)

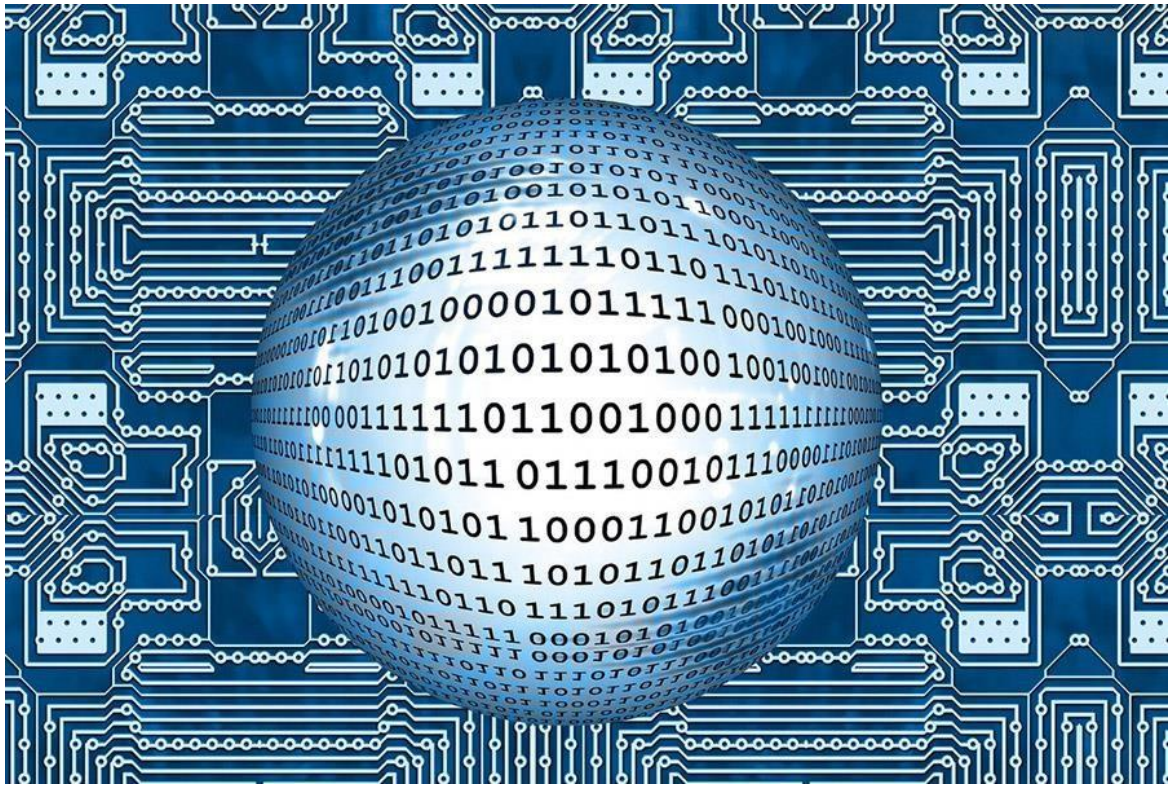




Study of discrete structures

- Core concepts in Computer Science: incorporates sets theory, graph, combinatorics, and algorithm,
- Has broader relevance and interdisciplinary application in cybersecurity, Natural Language Processing, Bioinformatics

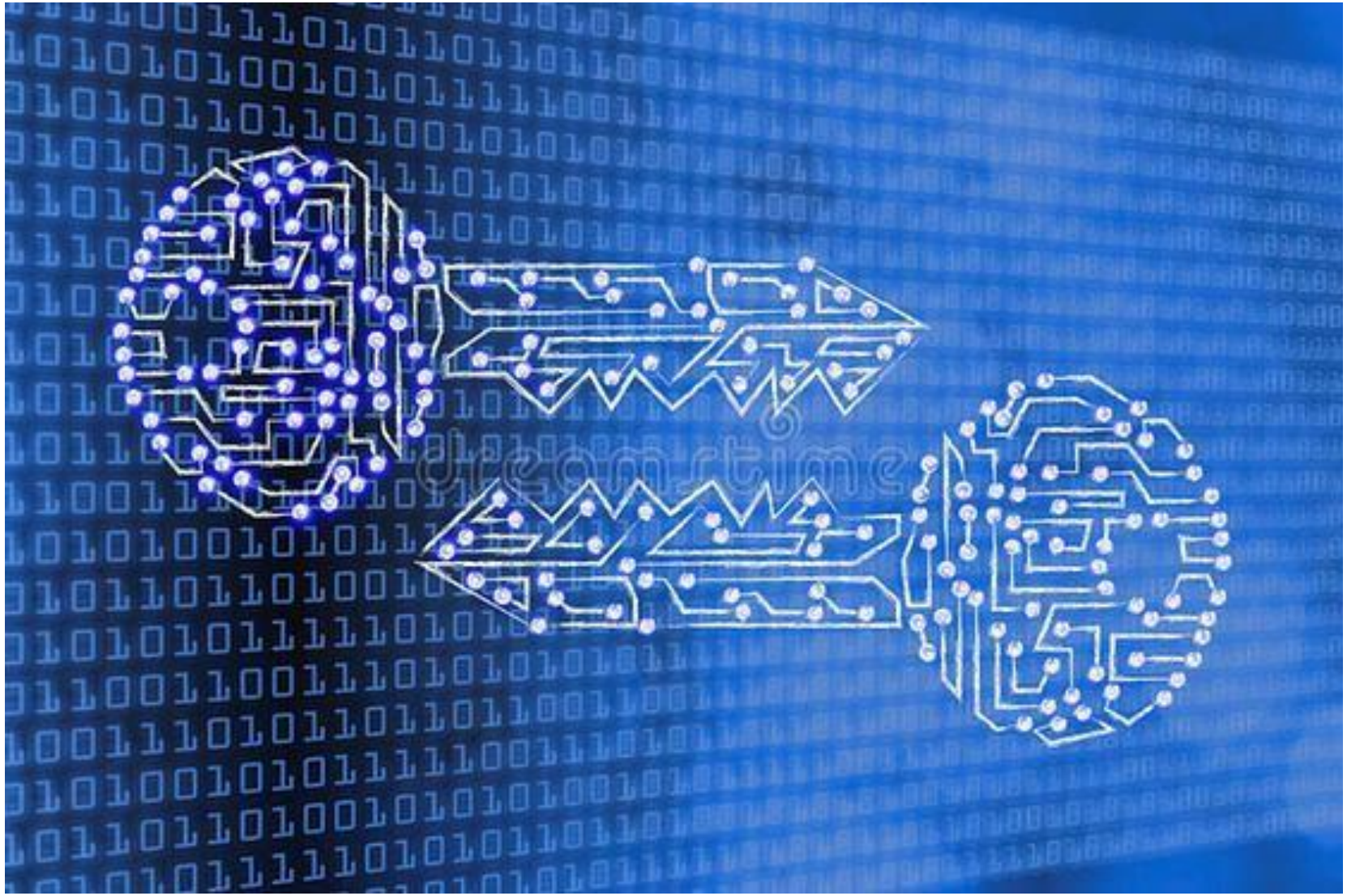
Few applications: discrete structure in computer storage



- Binary number system
  - The binary system uses only 0s and 1s to represent data in computers. This is a discrete structure because it involves separate, distinct value

Few applications: Cryptography

- Number theory
- Graph theory
- Combinatorics
- Symmetric and asymmetric encryption



Few applications: Cellular communication

- Digital signal processing
- Network protocol
- Channel encoding



Few applications: Algorithms in AI, Banking and Finance

TECH

A.I. has a discrimination problem. In banking, the consequences can be severe

PUBLISHED FRI, JUN 23 2023•1:45 AM EDT | UPDATED FRI, JUN 23 2023•10:37 AM EDT

CNBC



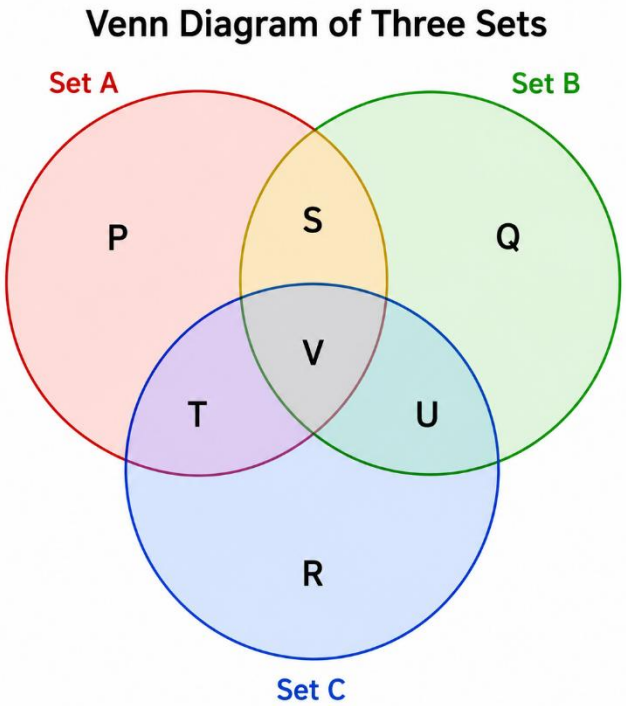
Artificial intelligence algorithms are increasingly being used in financial services — but they come with some serious risks around discrimination.

Chapter 1: Fundamentals

Introduction to Set Theory

What is set?

- A set is a collection of well-defined, distinct objects
- Few Examples:
  - {1, 2, 3}
  - {b, y, t, e}
  - {1, 3, 2, 3, 1} (same as {1, 2, 3})



Notation and terminology

- Sets are denoted by uppercase letters (e.g., A, B, C).
- Elements are denoted by lowercase letters (e.g., a, b, c).
- Membership:
  - $x \in A$ (x is an element of A)
  - $x \notin A$ (x is not an element of A).
- Example:
  - A = {1, 2, 3, a, b}
  - $1 \in A$
  - $5 \notin A$
- empty set { } or ϕ
- equal sets A=B. A = {1, 2, 3}, B = {2, 1, 3}, then A = B

Defining sets by properties

- Use the notation P(x) to represent a condition or property involving the variable x. The set defined by P(x), written as {x | P(x)}, consists of all elements x that satisfy the condition P(x).
- Z^+ = {x | x is a positive integer}, Z^+ = {1, 2, 3,…}
  - Example:
  - define a set of all even numbers less than 10.
  - **Condition**: The element x should be an even number less than 10.
  - **Notation:** {x | x is an even number and x < 10}
  - Set = {2, 4, 6, 8}
- The element x should be a letter that appears in the word “discrete".
  - Notation: {x | x is a letter in the word “discrete"}
  - Set = {d, i, s, c, r, e, t}

Subset and equality

- Subset
  - $A \subseteq B$ (A is a subset of B)
  - $A \nsubseteq B$ (A is not a subset of B)
- Equality
  - Two sets A and B are equal if they have the same elements: A = B
- Example
  - $Z^+ \subseteq Z$ (Z represents all integer numbers)
  - If $A = \{1, 2, 3\}$ and $B = \{x \mid x \text{ is a positive integer and } x^2 < 12\}$, then $A = B$.

Finite and infinite set

- Finite sets
  - Sets with a countable number of elements.
  - Example: {1,2,3}
- Infinite sets
  - Sets with an uncountable number of elements
  - Example: Z (all integers)

Cardinality of sets

- Cardinality measure of the "size" of the set. For finite sets, the cardinality is simply the count of distinct elements within the set.
- If A={1,2,3}, the cardinality of A, denoted as |A|, is 3

Exercises

1. Let $A = \{1, 2, 4, a, b, c\}$. Identify each of the following as true or false.

(a) $2 \in A$ **(b)** $3 \in A$ **(c)** $c \notin A$

(d) $\varnothing \in A$ **(e)** $\{\ \} \notin A$ **(f)** $A \in A$

- Answer

a. True

b. False

c. False

d. False

e. True

f. False

Exercises

2. Let $A = \{x \mid x \text{ is a real number and } x < 6\}$. Identify each of the following as true or false.

(a) $3 \in A$ **(b)** $6 \in A$ **(c)** $5 \notin A$

(d) $8 \notin A$ **(e)** $-8 \in A$ **(f)** $3.4 \notin A$

- Answer

a. True

b. False

c. False

d. True

e. True

f. False

Exercises

4. Give the set by listing its elements.

(a) The set of all positive integers that are less than ten.

(b) $\{x \mid x \in \mathbb{Z} \text{ and } x^2 < 12\}$

- Answer

a. {1, 2, 3, 4, 5, 6, 7, 8, 9}

b. {-1, -2, -3, 0, 1, 2, 3}

Exercises

5. Let $A = \{1, \{2, 3\}, 4\}$. Identify each of the following as true or false.

(a) $3 \in A$ **(b)** $\{1, 4\} \subseteq A$ **(c)** $\{2, 3\} \subseteq A$

(d) $\{2, 3\} \in A$ **(e)** $\{4\} \in A$ **(f)** $\{1, 2, 3\} \subseteq A$

- Answer

a. False

b. True

c. False

d. True

e. False

f. False

Exercises

write the set in the form {x | P(x)}, where P(x) is a property that describes the elements of the set
{2, 4, 6, 8, 10}

- x | x is $2 \leq x \leq 10$ (even positive integer)

Take home task

- Read Chapter 1